TEXORA · LECTURE SHEET

Moisture Content and Moisture Regain

Every textile fibre trades water with the air. Regain and content are the two ways of stating how much water is present — and because fibre is bought and sold by weight, the difference between them is measured in money.

TE-3103 · QUALITY EVALUATION OF FIBRE AND YARN · TOPIC 2 · LEVEL 3 TERM 1

1. Definitions that must never be confused

Let W = mass of water in a specimen, D = oven-dry mass of the same specimen, and W + D = its total (wet) mass.

$$\text{Moisture regain, } R\ \% = (W / D) \times 100$$

$$\text{Moisture content, } M\ \% = (W / (W + D)) \times 100$$

Regain expresses the water as a percentage of the *dry* material; **content** expresses it as a percentage of the *total* material. Regain is the commercial and technical convention in textiles because dry mass is the constant quantity — the amount of actual fibre does not change when the weather does. The two are interconvertible:

$$M = 100R / (100 + R) \quad \cdot \quad R = 100M / (100 - M)$$

The **oven-dry mass** is the mass after drying to constant mass under the standard conditions for the fibre concerned — for most fibres in a ventilated oven at 105 °C (commonly specified as 105–110 °C), weighing in a stoppered bottle so the hot, thirsty specimen cannot re-absorb moisture from the room air while it is on the balance.

WORKED EXAMPLE — REGAIN AND CONTENT FROM ONE WEIGHING

A cotton specimen weighs 108.5 g as received and 100.0 g oven-dry. Water W = 108.5 − 100.0 = 8.5 g.

Regain R = 8.5 / 100.0 × 100 = **8.5 %**. Content M = 8.5 / 108.5 × 100 = **7.83 %**.

Check by conversion: M = 100 × 8.5 / (100 + 8.5) = 850 / 108.5 = 7.83 % ✓. Note that regain is always the larger number, because its denominator is smaller.

2. Why fibres hold water — absorption and the regain curve

Textile fibres are polymers carrying polar groups (—OH in cellulose, —CONH— and —NH_2/—COOH in proteins) that attract water molecules. Water is held first as a tightly bound layer on these sites, then in progressively looser layers and finally in fine capillaries. Consequences:

- At a fixed temperature, regain rises with **relative humidity (RH)** in an S-shaped (sigmoid) curve — slowly at first, faster in the middle range, steeply above about 80 % RH as capillary water is added.
- **Hysteresis.** The desorption curve (drying down) lies above the absorption curve (wetting up): at 65 % RH cotton holds measurably more water when it arrives from a damp state than when it arrives from a dry state. This is why conditioning direction is specified (Topic 1).
- Absorption releases heat (**heat of sorption**) and causes **swelling** — mainly across the fibre (cotton swells about 20 % in cross-sectional area when wet, very little in length), which changes fabric dimensions and cover.
- Moisture changes properties: mass, strength (cotton and flax are *stronger* wet; viscose much *weaker* wet), extensibility, electrical resistance (static problems appear at low regain), and dyeing behaviour.

FIBRE	STANDARD (COMMERCIAL) REGAIN %	REGAIN AT 65 % RH, 20 °C (APPROX. %)
Cotton	8.5	7–8.5
Flax / linen	12	≈ 10–12
Jute	13.75	≈ 12–14
Wool	16 (some trades 14–18 by form)	≈ 14–16
Silk	11	≈ 10–11
Viscose	13	≈ 11–13
Acetate	6	≈ 6
Nylon 6.6	4.5	≈ 4
Polyester	0.4 (trade allowance often 0.4–1.5)	≈ 0.4
Acrylic	2.0	≈ 1.5–2

The **standard regain** (also called the commercial or official regain) is the agreed value used to compute the **correct invoice mass** of a consignment regardless of the weather on delivery day — see §4.

3. Measurement — the oven (gravimetric) method

The reference method for regain and content:

1. Draw the sample rapidly and seal it in an airtight container at once — a cotton sample left on a bench changes regain while it waits.
2. Weigh the specimen (typically 5–50 g, in duplicate) — mass as received.
3. Dry in a ventilated **conditioning oven** at 105–110 °C (fibre-dependent; heat-sensitive fibres use lower temperatures or vacuum drying as their standard directs) until constant mass — successive weighings differ by less than 0.05 % (or the limit the standard states).
4. Weigh the dry specimen in its **stoppered weighing bottle**, dried with it, so no moisture is regained during weighing. Conditioning ovens with an internal balance (Figure 1) remove even this transfer.
5. Calculate R % and, if required, M % from the two masses.

Faster routine methods — resistance and capacitance moisture meters, microwave and infrared balances, and the Shirley moisture meter — are all secondary: they respond to a property that changes with moisture and must be calibrated against the oven method for each fibre and form. They are excellent for process checks and misleading when treated as absolute.

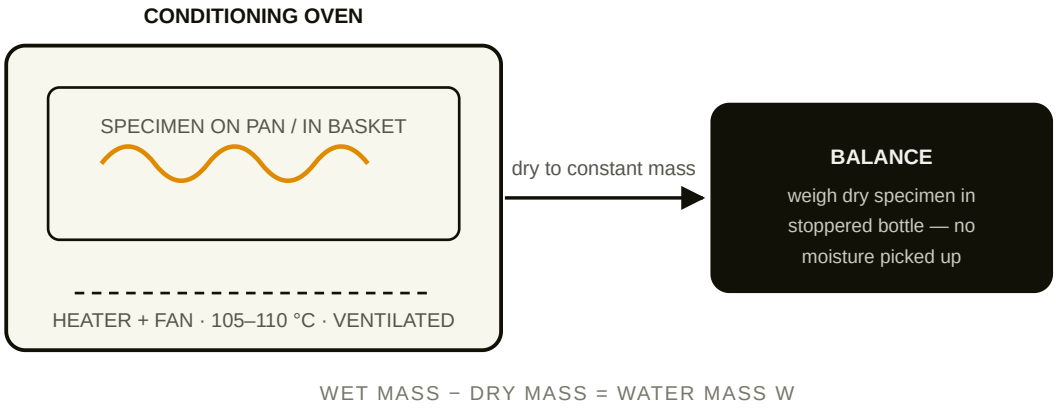


Figure 1 — Oven drying and weighing arrangement. The stoppered bottle (or an in-oven balance) is essential: a hot dry specimen re-absorbs moisture within minutes in room air.

4. Commercial use — correct invoice mass

Fibre and yarn are sold by mass, but the fibre — not the weather — is what the buyer wants. The **correct invoice mass** is the oven-dry mass increased by the standard regain:

$$\text{Correct invoice mass} = D \times (1 + R_{\text{standard}} / 100)$$

WORKED EXAMPLE — INVOICE MASS OF A COTTON CONSIGNMENT

A consignment arrives with a gross fibre mass of 200.0 kg at the regain found on delivery. Testing gives an oven-dry mass (from representative samples, scaled to the consignment) of 184.33 kg.

> Cotton's standard regain is 8.5 %.
> Correct invoice mass = 184.33 × 1.085 = **200.0 kg** — the delivery happened to be exactly at standard regain, so the invoiced and delivered masses agree.
> Had the same dry fibre arrived wetter, say at 11 % actual regain, its delivered mass would have been 184.33 × 1.11 = 204.6 kg — but the invoice mass would still be 200.0 kg. The buyer does not pay for the extra 4.6 kg of water. Conversely a bone-dry delivery is credited up to the same 200.0 kg. This is why regain testing is done on samples drawn at the moment of weighing, jointly where large sums are involved.

The same principle corrects yarn count: a count measured on yarn at an unknown regain is corrected to the count at standard regain before it is compared with the nominal count (used again in Topic 10).

5. Sources of error

- **Sampling delay / unsealed containers** — the sample equilibrates with the laboratory air, not with the lot.
- **Incomplete drying** — stopping before constant mass overstates dry mass and understates regain.
- **Over-drying / degradation** — excessive temperature or time drives off non-water volatiles (oils, finishes) and can degrade the fibre, understating dry mass. Hence fibre-specific drying conditions.
- **Weighing an unstoppered hot specimen** — rapid re-absorption plus convection currents around the warm bottle falsify the dry mass.
- **Finishes and oils** — the oven method reports all mass lost as "water"; extracted finishes inflate the apparent regain unless the standard's extraction step is applied.

Glossary

Moisture regain — water mass as a percentage of oven-dry mass. **Moisture content** — water mass as a percentage of total mass. **Oven-dry mass** — mass after drying to constant mass under specified conditions. **Standard (commercial) regain** — the agreed regain used for invoice mass; cotton 8.5 %. **Correct invoice mass** — dry mass × (1 + standard regain/100). **Hysteresis** — absorption and desorption curves differ; regain at a given RH depends on the direction of approach. **Conditioning oven** — a ventilated oven, often with internal balance, used for drying specimens to constant mass.

Key points

• R % is based on dry mass, M % on total mass; R is always larger; M = 100R/(100+R).
• Cotton's standard regain is 8.5 %; wool's is about 16 %; polyester's about 0.4 % — the fibre's chemistry sets the value.

• The oven method (105–110 °C, constant mass, stoppered weighing) is the reference; all meters are calibrated against it.
• Correct invoice mass removes the weather from trade: dry mass brought up by the standard regain.
• Hysteresis means regain depends on history — control the approach direction when conditioning.

Practice

1. A wool specimen weighs 58.0 g as received and 50.0 g oven-dry. Find the regain and the content.
2. A yarn has a moisture content of 6.0 %. What is its regain?
3. Samples from a jute consignment give an oven-dry mass of 880 kg for the lot. Using the standard regain of jute (13.75 %), find the correct invoice mass.
4. Why is regain, not content, the basis of textile trade? Give the practical reason.
5. A technician dries cotton at 140 °C "to save time" and weighs the specimen unstoppered after it has stood 10 minutes on the bench. State the direction of each error's effect on the reported regain.

Answer key

1. W = 8.0 g. R = 8.0/50.0 × 100 = **16.0 %** (wool's standard regain, as expected). M = 8.0/58.0 × 100 = **13.79 %**.
2. R = 100M/(100 − M) = 600/94 = **6.38 %**.
3. Invoice mass = 880 × 1.1375 = **1001.0 kg**.
4. Dry mass is the invariant quantity of actual fibre; total mass changes with the weather. Basing percentages and invoice mass on dry mass means both parties price the same thing — the fibre — whatever the delivery-day humidity.
5. Overheating drives off non-water volatiles and may degrade the fibre → dry mass too low → regain **overstated**. Standing unstoppered, the dry specimen re-absorbs room moisture → dry mass too high → regain **understated**. The two errors partly cancel by luck, which is worse than either alone because the result looks plausible while being wrong.